

1. Gray Level Transformation:

a. purpose.

Gray level transformation is used to enhance image quality by changing pixel intensities to improve visibility and contrast.

b. Negative Transformation

formula: $S = 255 - r$.

Input r	output s
50	205
100	155
150	105
200	55

Interpretation : Bridge areas become dark and areas become bright, producing a negative image.

2. Histogram equalisation

Histogram

Intensity	frequency.
0	4
1	6
2	10
3	6

Total pixels = 26

a. purpose.

histogram equalization improves image contrast by spreading pixel intensities over the available range.

b.

CPF

Intensity	frequency	CPF
0	4	4
1	6	10
2	10	20
3	6	26

c. Equalised Intensities.

Intensity	Equalised value.
0	0
1	1
2	2
3	3

$$s = \frac{\text{CPF}}{26} \times 3.$$

Interpretation: The contrast becomes more uniform making image details easier to distinguish.

3. Gaussian smoothing.

a. what why Gaussian filtering?

Gaussian filtering reduces noise smoothly and preserving edges better than averaging filtering

b) calculate:

$$\text{kernel: } \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{image: } \begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

new center value:

$$= \frac{1}{16} [10 + 40 + 10 + 40 + 160 + 160 + 10 + 40 + 10]$$

$$= \frac{360}{16}$$

$$= 22.5$$

$$= 23.$$

Interpretation:

Noise is reduced and the image becomes smoother.

h. Edge Detection using Sobel operator.

a) purpose.

Edge detection identifies object boundaries by detecting sudden intensity changes.

b) horizontal sobel.

$$\text{mask: } \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{image: } \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Gradient

$$= (-10 -20 -10) + 0 + (30 +60 +30)$$

$$= 80$$

$$\boxed{\text{Gradient} = 80}$$

Interpretation: A strong horizontal edge is detected.

5. Contrast stretching.

a) need.

*1) Contrast stretching expands the intensity range making the image cleaner and improving visibility.

b) linear stretching.
formula,

$$s = \frac{(r-50)}{150-50} \times 255.$$

for $r=50$: $s=0$

for $r=100$;

$$s = \frac{50}{100} \times 255 = 127.5 = 128.$$

for $r=150$;

$$s = 255.$$

Input	output:
50	0
100	128
150	255

Interpretation: The intensity range is expanded from 50-150 to 0-255 - resulting in a high-contrast and more detailed image.